

Hormone changes across the menstrual cycle

How do hormone graphs reveal the control sequence behind the cycle?

Learning goal

Interpret FSH, LH, estrogen and progesterone patterns and explain the switch from negative to positive feedback before ovulation.

Before you begin

A graph shows a variable against an axis. Check what is measured before comparing the height of different curves.

How to complete this lesson Learn → worked example → practise → lesson check

- Read the explanation and its diagrams. Open a bold term when you need a definition.
- Follow the worked example, then try both supported questions. Select Check answer for feedback.
- Open the required check and select Start. Correct both selections to unlock writing.
- Write each explanation and select its Save written response button. Editing saved writing makes it a draft again.
- Select Submit and finish to complete the check. Find your records in All My Work.

Expected work: one completed lesson check. Supported practice, videos and textbook questions are optional. Saved writing is not automatically graded.

Chapter 14 · printed pp. 495–502

Open p. 495 · Open p. 498 · Open p. 500 · Open p. 502

Key terms: estrogen · progesterone · LH surge · negative feedback

Vocabulary help

Key terms

Words for this lesson

Use these definitions when you need help with a term.

estrogen

An ovarian hormone that supports endometrial growth and participates in feedback.

progesterone

A hormone that supports and maintains the endometrium after ovulation.

LH surge

A large, brief increase in LH release associated with ovulation.

negative feedback

A response that tends to oppose the initial change.

More vocabulary for this lesson

Lesson vocabulary

positive feedback

A response that reinforces or amplifies a change.

Name the hormone source before interpreting a curve

A hormone curve records how a quantity changes over time; it does not show the producing gland directly. Before following its peaks, connect each hormone with its source and job. GnRH, gonadotropin-releasing hormone, comes from the hypothalamus and stimulates the anterior pituitary to release FSH and LH.

FSH, follicle-stimulating hormone, supports follicular development in the ovary. LH, luteinizing hormone, contributes to ovarian activity and its surge helps trigger ovulation. These are pituitary signals arriving at ovarian tissue. Estrogen and progesterone in this cycle are outputs of ovarian tissue rather than hormones made by the anterior pituitary.

As the dominant follicle develops, its estrogen output increases. Estrogen supports endometrial growth. Through much of the follicular phase, estrogen also contributes to negative feedback that limits further pituitary stimulation, especially FSH. Negative feedback is a response that tends to oppose the original change; it does not mean the hormone is harmful.

Keep direct and later effects separate. FSH supports a developing follicle, the follicle produces estrogen, and estrogen affects the uterine lining. Calling all of these 'female hormones' would hide the source-to-target links that explain a graph.

Sustained high estrogen precedes the LH surge

Near the end of follicular development, sufficiently high estrogen sustained over time changes the feedback response. Instead of restraining the control system, this ovarian signal promotes a strong pituitary LH surge: a large, brief rise in LH release. Positive feedback means that a response reinforces or amplifies activity in the pathway.

The LH surge helps trigger release of the secondary oocyte and the changes that form the corpus luteum from remaining follicular tissue. FSH also has a smaller rise around this event. The large, narrow LH peak is therefore a useful landmark when reading the normal model, but it is a hormone event linked to ovulation, not a picture of the oocyte itself.

The positive-feedback interval is temporary and depends on the stage and sustained signal. It does not mean every estrogen increase always raises LH, that all inhibition disappears permanently, or that hormone levels become fixed after the surge. 'Estrogen inhibits the

pituitary' and 'estrogen helps produce an LH surge' describe different conditions in the same cycle.

To explain the switch, state both conditions and direction: ordinary follicular feedback helps restrain stimulation, whereas sustained high pre-ovulatory estrogen promotes the surge. A single measurement without its timing and pattern does not establish which response is occurring.

Follow the post-ovulatory ovarian output

After ovulation, an active corpus luteum releases progesterone and some estrogen. Progesterone rises and supports the endometrium. The luteal hormones also contribute to negative feedback on the hypothalamic-pituitary system, helping limit the stimulation that would support another ovulatory event during this phase.

Without pregnancy support, the corpus luteum regresses. Its progesterone and estrogen output falls, reducing support for the lining and contributing to the next menstrual flow. A broad progesterone rise after ovulation fits the change from follicle to corpus luteum; it should not be placed as the cause of the earlier follicle's initial growth.

Estrogen can show a smaller rise during the luteal phase as well as the larger pre-ovulatory rise. Read the whole pattern rather than insisting that a hormone must come from only one ovarian structure or have only one peak. These sketches simplify a continuous process and do not show every regulating hormone.

Read timing and scale from the actual graph

Begin with the axes. Both panels in the retained figure use cycle day horizontally, with day 1 marking the start of menstrual flow in this illustration. The vertical axes say relative level. There are no measured blood-concentration units or numerical vertical ticks. Each curve is separately scaled, so a line twice as high is not evidence of twice the hormone concentration.

Read the key with line pattern as well as colour. In the upper panel, solid reddish LH has the narrow high peak near day 14; dashed brown FSH has a smaller rise near that event and a modest earlier rise. In the lower panel, solid teal estrogen rises before the large LH peak and has a smaller later rise. Dashed dark-green progesterone rises later, with a broad maximum near the third week of the illustrated cycle.

Compare the curves against the same day marks: the estrogen rise precedes the LH surge, and the broad progesterone rise follows it. Then connect each to biology: developing follicle, pituitary surge/ovulation, active corpus luteum. The line segment close to the bottom is a low relative level in this schematic; it does not prove that a hormone is completely absent.

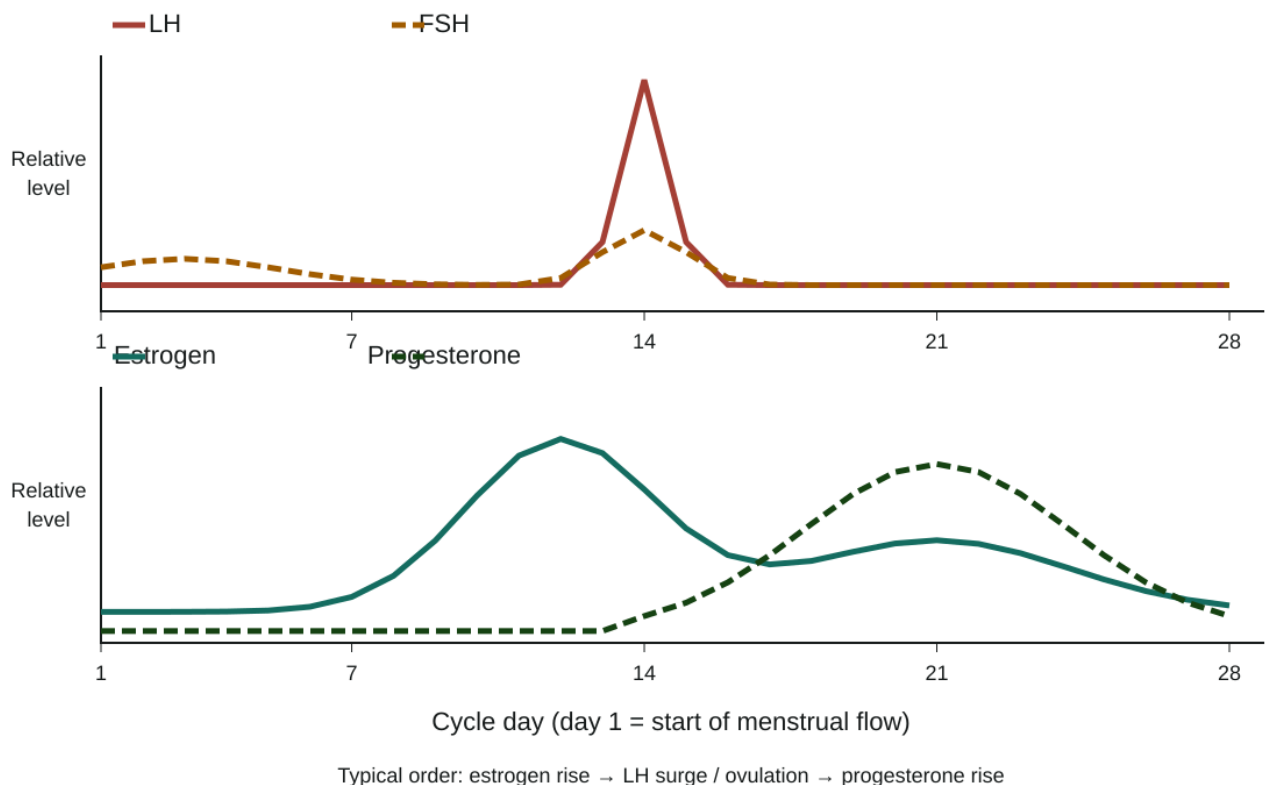
Curve height, concentration and rate of change are different comparisons. Timing of peaks can be compared here because the panels share the day axis. Change within one curve can be described qualitatively as rising or falling. Absolute concentration ratios and numerical rates cannot be calculated from this figure because the vertical scale supplies no measured values.

For a new graph, inspect its own units, vertical ranges, labels and stated conditions. A different colour is not a different hormone, and a shifted peak should not be forced onto day 14. Identify the observed change, connect it to the model, and state what the record does not establish. These teaching graphs are not individual observations or a personal fertility calendar.

An observation record also has a sampling limit. A brief peak might fall between measurements or outside the recorded interval. A missing recorded peak therefore does not by itself establish that the biological event never happened; first distinguish what was measured from what remains unobserved.

Hormone patterns across an illustrative cycle

Schematic relative levels — not measured data; compare timing, not absolute amounts



An authored qualitative model based on the teaching sequence. Curves are separately scaled and are not patient observations.

Worked example

A taller trace can represent a smaller measured value

Consider a constructed scale-reading exercise, not patient data. Graph A has a vertical range from 0 to 20 units/L; its marked point is three-quarters of the way up. Graph B has the same displayed height but a vertical range from 0 to 200 units/L; its point is halfway up. Both use linear vertical scales and the same stated unit. Which point has the larger numerical value?

- Read the range before comparing height. Graph A's full height represents 20 units/L, while Graph B's full height represents 200 units/L. Equal centimetres on the page do not represent equal amounts.

- Convert each position using its own axis. Three-quarters of 20 is 15 units/L. Half of 200 is 100 units/L. The linear scales begin at zero in this exercise, so these fractions apply directly to the full ranges.
- Compare the values in their common unit: 100 units/L is greater than 15 units/L, even though Graph B's point is drawn lower. The naive height comparison gives the wrong conclusion.
- Return to the retained cycle schematic. It does not provide these numerical ranges at all, so we cannot perform the same conversion or calculate concentration ratios from it. We can still compare the timing of rises and peaks along its day axis.
- Keep the example bounded. The invented numbers demonstrate axis reading; they are not normal ranges for LH, FSH, estrogen or progesterone, and do not support a diagnosis.

Optional application

A hypothetical record meets the model's sustained-high-estrogen condition but shows no recorded LH surge in the following observation interval. Which expected relationship is missing from that record? Explain why this alone does not prove a particular cause or establish that ovulation never occurred.

Answer withheld for first attempt

Practise with support

Apply the explanation

Try both questions. Use the hint to revise your thinking after an incorrect answer.

What change most directly precedes the LH surge in the normal model?

- A fall in all ovarian hormones
- Regression of the corpus luteum at the end of the cycle
- Sustained high estrogen from the developing follicle

A broad progesterone rise occurs after the LH peak. What ovarian change explains it?

- Formation of an active corpus luteum
- Loss of every follicular cell from the ovary
- The oocyte becoming a hormone-producing gland

Stop and think

Try this before opening the explanation

Is estrogen's feedback always negative?

Answer withheld for first attempt

Optional video support

Hormonal control of the cycle

As you watch: Trace the estrogen rise, LH surge and later progesterone rise.

Optional video. Internet access is required. The explanation below covers the idea without the video.

Read the explanation

Sustained high estrogen before ovulation promotes an LH surge. The surge helps trigger ovulation. The corpus luteum then releases progesterone; its later regression allows hormone support to fall.

Open required check · Hormone changes across the menstrual cycle

Explain what you have learned

This check counts toward chapter progress. Correct the 2 selections to unlock 2 written explanations. Saving writing records your response; it does not grade its accuracy.

Selection 1

Which hormone change most directly triggers ovulation in the usual cycle?

- Fall in all pituitary hormones
- Peak in inhibin alone
- LH surge
- Regression of the corpus luteum

Selection 2

Why is the pre-ovulatory estrogen-LH relationship an example of positive feedback?

- Progesterone has already caused menstruation
- The hormone concentrations cannot change after the surge
- Estrogen removes every inhibitory signal permanently
- High estrogen promotes increased pituitary stimulation leading to an LH surge

Answer the selections correctly to unlock the writing.

Written explanation 1

Explain the normal order of estrogen, LH and progesterone changes around ovulation. Name the relevant ovarian structures.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Written explanation 2

Two curves use different vertical scales. Explain why comparing their visual heights may be misleading and describe a comparison you can still make.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Optional application 2

This is optional written practice. It does not change required progress.

Explain the positive-feedback relationship leading to the LH surge before ovulation. Contrast it with estrogen's inhibitory contribution at other stages.

[Write your own response separately.]

Optional response. Select Save response to collect it. Maximum 5000 characters.

← Coordinating the ovarian and uterine cycles / Next: When hormone patterns change →